

> Previous Years' Questions

395. Cancer is more common in older people because **2013**

- (a) their immune systems have degenerated
- (b) the supply of certain hormones declines with age
- (c) their bodies are unable to adjust to the changing environment
- (d) they have accumulated more mutations

396. Consider the following statements regarding antibiotics. **2013 (I)**

- 1. They are used to destroy disease-causing bacteria.
- 2. They can be applied to the skin, swallowed or injected to fight microorganisms inside the body.
- 3. They are effective against disease-causing viruses.
- 4. The first antibiotic to be discovered was tetracycline.

Which of the statement(s) given above is/are correct?

- (a) 1 and 2
- (b) 1, 2 and 4
- (c) Only 1
- (d) 2, 3 and 4

397. Which one of the following diseases in humans can spread through air? **2015 (I)**

- (a) Dengue
- (b) Tuberculosis
- (c) HIV-AIDS
- (d) Goitre

398. The word 'vaccination' has been derived from a Latin word which relates to **2015 (I)**

- (a) pig
- (b) horse
- (c) cow
- (d) dog

399. Measles is a disease caused by **2016**

- (a) bacteria
- (b) virus
- (c) protozoan
- (d) worm

400. The mandate of the scheme entitled 'Directly Observed Treatment, Short-Course (DOTS)' launched by WHO is to ensure that

- (a) doctors treat patients with medicine for a short duration
- (b) doctors do not start treatment without a trial
- (c) patients complete their course of drug
- (d) patients voluntarily take vaccines

401. Penicillin inhibits synthesis of bacterial **2016 (I)**

- (a) cell wall
- (b) protein
- (c) RNA
- (d) DNA

402. Most antibiotics target bacterial parasites interfering with various factors of growth or metabolism such as **2016 (I)**

- 1. synthesis of cell wall.
- 2. bacterial protein synthesis.
- 3. synthesis of nuclear membrane.
- 4. mitochondria function.

Select the correct answer using the codes given below:

- (a) 1, 2 and 3
- (b) 1 and 4
- (c) 2 and 3
- (d) 1 and 2

403. Dengue virus is known to cause low platelet count in blood of patient by **2016 (I)**

- 1. interfering in the process of platelet production in bone marrow.
- 2. infecting endothelial cells.
- 3. binding with platelets.
- 4. accumulating platelets in intestine.

Select the correct answer using the codes given below:

- (a) 1 and 2
- (b) 1 and 3
- (c) 3 and 4
- (d) 1, 2 and 3

Applied Biology

404. If excess fertiliser is applied to a plant without water, the plant will

- (a) be stunted in growth
- (b) develop modifications
- (c) die due to plasmolysis
- (d) remain unaffected

405. The free-living aerobic soil bacterium which fixes nitrogen

- (a) *Azotobacter*
- (b) *Nostoc*
- (c) *Clostridium*
- (d) *Frankia*

406. Match the following Columns.

Column I	Column II
A. Sugarcane	1. Quinine
B. <i>Cinchona</i>	2. Molasses
C. <i>Hevea</i>	3. Iodine
D. Sea kelp	4. Rubber

Codes

- | | |
|-------------|-------------|
| A B C D | A B C D |
| (a) 2 1 4 3 | (b) 4 3 2 1 |
| (c) 1 2 3 4 | (d) 2 3 4 1 |

407. Consider the following microorganisms.

- 1. *Azotobacter*
- 2. *Clostridium*
- 3. *Bacillus polymyxa*
- 4. VAM (Vascular Arbuscular Mycorrhiza)

Which of the microorganism(s) given above is/are free-living nitrogen-fixing bacteria?

- (a) Only 1
- (b) 1, 2 and 3
- (c) Only 3
- (d) 2, 3 and 4

408. Consider the following plants.

- 1. Lentil
- 2. Cow pea
- 3. Sunnhemp
- 4. Wheat

Which of the plants given above is/are used as 'Green manure in India'?

- (a) Only 3
- (b) 1, 2 and 3
- (c) 2 and 3
- (d) All of these

409. What are the components of the fungicide 'Bordeaux mixture'?

- (a) Magnesium sulphate and sodium hydroxide
- (b) Copper sulphate and magnesium hydroxide
- (c) Copper sulphate and calcium hydroxide
- (d) Copper sulphate and sodium hydroxide

410. In the rice field, which of the following is used as biofertiliser?

- (a) *Rhizobium*
- (b) *Pseudomonas*
- (c) *Nitrobacter*
- (d) *Azolla*

411. From which plant listed below maximum amount of ethanol for biofuel is obtained?

- (a) Wood
- (b) Tapioca
- (c) Maize
- (d) Sugarcane

412. Plants die when over fertilised, because the fertiliser

- (a) damage the wall of delicate root hairs
- (b) blocks absorption of nitrogenous ions
- (c) causes dehydration of plant by exosmosis
- (d) increases the soil acidity

413. Which one among the following group of items contains only biodegradable items?

- (a) Wood, grass, plastic
- (b) Wood, grass, leather
- (c) Fruit peels, lime juice, China clay cup
- (d) Lime juice, grass, polystyrene cup

414. Endosulfan, which has been in news these days, is a/an

- (a) pesticide
- (b) fertiliser
- (c) sulpha drug
- (d) antibiotic

415. Which one among the following industries produces the most non-biodegradable wastes?

- (a) Thermal power plants
- (b) Food processing units
- (c) Textile mills
- (d) Paper mills

416. Which among the following statements about biofertilisers are correct?

1. *Azotobacter* is one of the nitrogen-fixing bacteria used as a biofertiliser.
2. They have to be applied to the leaves of the plant only.
3. They alter the chemical composition of the soil.
4. They can be used along with organic fertilisers.

Select the correct answer using the codes given below.

- (a) 1 and 4 (b) 1, 2 and 4
(c) 2 and 3 (d) All of these

417. Which one of the following is not biodegradable?

- (a) Woollen mat (b) Silver foil
(c) Leather bag (d) Jute basket

418. Which one of the following plants is used for green manuring in India?

- (a) Wheat (b) Sunnhemp
(c) Cotton (d) Rice

419. Which among the following are the most important raw materials for manufacturing of soap?

- (a) Fats and caustic alkali
(b) Fats and potash
(c) Fats and acid
(d) Vegetable oil and potash

420. Consider the following statements about Vechur cattle breed.

1. Vechur is the world's smallest cow.
2. It is indigenous breed found in Kerala.
3. Its milk protein has medicinal value.
4. Commonly used in farming.

Which of the statements given above are correct?

- (a) 3 and 4 (b) 1, 2 and 3
(c) 2 and 3 (d) All of these

421. Cattle are capable of digesting cellulose present in the grass and/or fodder that they eat. This ability is attributed to the

- (a) presence of cellulose degrading bacteria in the rumen
(b) production of cellulose by the cattle rumen
(c) acids present in the rumen
(d) prolonged retention of cellulose in the rumen

422. ELISA test is prescribed for

- (a) cancer (b) typhoid
(c) polio (d) AIDS

423. Restriction enzymes are those enzymes that can

- (a) cut RNA
(b) cut single-stranded DNA
(c) cut double-stranded DNA
(d) hydrolyse proteins

424. Which one among the following is the 'chemical knives (scissors)' used in genetic engineering?

- (a) Polymerase (b) Ligases
(c) Endonucleases (d) Plasmid

425. Match the following Columns.

Column I (Techniques)	Column II (Compounds)
A. Northern blotting	1. RNAs
B. Southern blotting	2. Proteins
C. Western blotting	3. Lipids
D. Far-Eastern blotting	4. DNAs

Codes

- A B C D A B C D
(a) 2 1 3 4 (b) 2 3 1 4
(c) 1 2 4 3 (d) 1 4 2 3

426. Match the following Columns.

Column I (Scientists)	Column II (Discoveries)
A. Banting and Best	1. Hybridoma technology
B. Milstein Kohler	2. Isolation of insulin
C. Steward Miller	3. <i>In vitro</i> fertilisation
D. PC Steptoe	4. Coconut milk in tissue culture

Codes

- A B C D A B C D
(a) 2 1 3 4 (b) 2 1 4 3
(c) 4 2 1 3 (d) 4 1 2 3

427. By using which one of the following techniques, is DNA fingerprinting done?

- (a) ELISA (b) RIA
(c) Northern blotting (d) Southern blotting

428. Which of the following techniques can be used to establish the paternity of a child?

- (a) Protein analysis
(b) Chromosome counting
(c) Quantitative analysis of DNA
(d) DNA fingerprinting

429. Which among the following is the first hormone produced by genetic engineering?

- (a) Insulin (b) Cortisol
(c) Thyroxine (d) Testosterone

430. The genetically engineered 'Golden rice' is rich in which of the following?

- (a) Vitamin-A and nicotinic acid
(b) β -carotene and folic acid
(c) β -carotene and iron
(d) Vitamin-A and niacin

431. Consider the following statements

1. Dolly, the cloned sheep.
2. George and Charlie, the cloned calves.
3. Headless frog.

Which of the statement(s) given above is/are incorrect?

- (a) Only 1 (b) 1 and 2
(c) Only 3 (d) All of these

432. Match the following Columns.

Column I	Column II
A. Maize	1. Arjun
B. Paddy	2. Jaya
C. Wheat	3. Ranjit

Codes

- A B C A B C
(a) 1 2 3 (b) 2 3 1
(c) 3 2 1 (d) 3 1 2

433. Which one of the following is responsible for converting milk into curd?

- (a) Fungi (b) Bacteria
(c) Virus (d) None of these

434. Which of the following is used in the production of yoghurt?

- (a) *Streptococcus thermophilus*
(b) *Lactobacillus bulgaris*
(c) *Streptococcus lactis*
(d) Both (a) and (b)

435. A milkman puts banana leaf in milk jar, because banana leaf

- (a) gives a fresh flavour to milk
(b) makes the milk acidic and resistant to yeast
(c) makes the milk basic and resistant to yeast
(d) increases the whiteness of milk

436. Match the following Columns.

Column I	Column II
A. Wine	1. Barley
B. Beer	2. Sugarcane juice
C. Whisky	3. Grapes
D. Rum	4. Molasses

Codes

- A B C D A B C D
(a) 2 1 4 3 (b) 3 4 1 2
(c) 3 1 4 2 (d) 2 4 1 3

437. Which one of the following is not a constituent of biogas?

- (a) Methane (b) Carbon dioxide
(c) Hydrogen (d) Nitrogen dioxide

438. Which of the following gases is released from rice fields in the most prominent quantities?

- (a) Carbon dioxide
- (b) Methane
- (c) Carbon monoxide
- (d) Sulphur dioxide

439. Consider the following statements about bioremediation.

1. It may be defined as any process that uses microorganisms or their enzymes to return the environment altered by contaminants to its original condition.
2. Bioremediation may be employed in order to attack specific contaminants, such as chlorinated pesticides that are degraded by bacteria.

Which of the statement(s) given above is/are correct?

- (a) Only 1
- (b) Only 2
- (c) 1 and 2
- (d) Neither 1 nor 2

440. Match the following Columns.

Column I (Microorganisms)	Column II (Uses)
A. <i>Aeromonas hydrophila</i>	1. Cloning vector
B. <i>Methanobacteria</i>	2. Microbial fuel cells
C. <i>Agrobacterium tumefaciens</i>	3. Yogurt preparation
D. <i>Lactobacillus bulgaricus</i>	4. Biogas synthesis

Codes

- | | |
|-------------|-------------|
| A B C D | A B C D |
| (a) 2 4 1 3 | (b) 2 1 4 3 |
| (c) 4 3 1 2 | (d) 4 1 3 2 |

441. Which among the following are the major reasons behind preferring *Eucalyptus* tree in the planned forestation process?

1. Plantation grows very fast.

2. Plantation makes the soil more fertile.

3. Wood from *Eucalyptus* tree is easily converted into pulp for paper industry.

Select the correct answer using the codes given below:

- (a) 1 and 2
- (b) 1 and 3
- (c) 2 and 3
- (d) All of these

> Previous Years' Questions

442. The macronutrients provided by inorganic fertilisers are

☞ 2012 (II)

- (a) carbon, iron and boron
- (b) magnesium, manganese and sulphur
- (c) magnesium, zinc and iron
- (d) Nitrogen phosphorus and potassium

443. Which one among the following groups of items contains only biodegradable items?

☞ 2013 (II)

- (a) Paper, grass, glass
- (b) Wood, flower, iron-scrap
- (c) Sewage, plastic, leather
- (d) Cowdung, paddy-husk, vegetable wastes

444. The main constituent of gobar gas or biogas is

☞ 2014 (I)

- (a) ethane
- (b) methane
- (c) propane
- (d) acetylene

445. Which one of the following types of pesticides is convenient to control stored grain pests?

☞ 2014 (II)

- (a) Systemic pesticides
- (b) Fumigants
- (c) Contact poisons
- (d) Stomach poisons

446. Which one among the following cattle breed produces highest amount of milk?

☞ 2014 (II)

- (a) Brown swiss
- (b) Holstein
- (c) Dutch belted
- (d) Blaarkop

447. Who among the following is considered as the 'father of genetic engineering'?

☞ 2016 (I)

- (a) Philip Drinker
- (b) Paul Berg
- (c) Thomas Addison
- (d) Alpheus S Packard Jr

448. Genetic screening is

☞ 2016 (I)

- (a) analysis of DNA to check the presence of a particular gene in a person
- (b) analysis of gene in a population
- (c) pedigree analysis
- (d) screening of infertility in parents

449. Genetically Modified (GM) crops contain modified genetic material due to the

☞ 2016 (I)

1. introduction of new DNA.
2. removal of existing DNA.
3. introduction of RNA.
4. introduction of new traits.

Select the correct answer using the codes given below:

- (a) 1 and 2
- (b) 1, 2 and 3
- (c) 3 and 4
- (d) 1, 2 and 4

450. Methyl isocyanate gas, which was involved in the disaster in Bhopal in December, 1984, was used in the Union Carbide factory for production of

☞ 2016 (I)

- (a) dyes
- (b) detergents
- (c) explosives
- (d) pesticides

451. The germplasm is required for the propagation of plants and animals. Germplasm is the

☞ 2016 (I)

1. genetic resources.
2. seeds or tissues for breeding.
3. egg and sperm repository.
4. a germ cell's determining zone.

Select the correct answer using the codes given below.

- (a) Only 1
- (b) 1, 2 and 3
- (c) 2 and 3
- (d) 2 and 4